

Physics of Diffusion Models in Generative AI

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Abstract

In this report, I explore Stochastic Differential Equations (SDEs) and their connection to Langevin dynamics. I examine what type of SDE Langevin dynamics correspond to in its undamped form. I then contrast how SDEs, Langevin dynamics and generative modeling are linked. Showing how the modern diffusion and score based models use Langevin type SDEs to generate data from noise.

1 Langevin dynamics

Langevin equations are a type of stochastic differential equations. As recently derived, the equation is given by:

$$\gamma \frac{dx}{dt} = -\nabla V + \sigma \zeta(t)$$

Taking the time average and simplifying, the solution to the differential equation becomes;

$$x(t) - x(0) = \frac{\sigma}{\gamma} \int_0^t \zeta(t') dt' \implies x(t) = x_0 + \frac{\sigma}{\gamma} \int_0^t \zeta(t') dt'$$

With initial condition $x(0) = x_0$

2 Stochastic Differential Equations (SDEs)

Stochastic differential equations are types of differential equations where change is combined with random noise. The solution of these differential equations are random, stochastic processes, rather than deterministic functions that evolve over time [3]. Consider an ordinary DE;

$$\frac{dx}{dt} = f(x, t) \implies dx = f(x, t) dt$$

The solution to the above DE;

$$x(t) = x(0) + \int_0^t f(x, t') dt'$$

If it is assumed a function $a(t)$ is not deterministic but stochastic, then there arises a stochastic differential equation, given by;

$$\frac{dx}{dt} = a(t)x(t) \tag{1}$$

where $a(t) = f(t) + h(t)\zeta(t)$, Thus, [2]

$$\frac{dX(t)}{dt} = f(t)X(t) + h(t)X(t)\zeta(t) \implies dX(t) = f(t)X(t)dt + h(t)X(t)\zeta(t)dt$$

where $\zeta(t)dt = dW(t)$ is the standard Wiener process. Thus,

$$dX(t) = f(t)X(t)dt + h(t)X(t)dW(t)$$

Therefore the general SDE, Ito sense is given by;

$$dX_t = \mu(X_t, t)dt + \sigma(X_t, t)dW_t \quad (2)$$

where

- X_t is the stochastic process
- $\mu(X_t, t)$ is the drift term, which has deterministic tendencies
- $\sigma(X_t, t)$ is the diffusion term, this term is the one that determines the noise intensity
- dW_t is an infinitesimal increment of Brownian motion

With solution

$$X(t) = X(0) + \int_0^t \mu(X_{t'}, t')dt' + \int_0^t \sigma(X_{t'}, t')dW_{t'}$$

3 How are SDEs linked to Langevin dynamics?

Langevin equations are physics motivated types of SDEs that describe particles subject to deterministic forces, friction and thermal noise.

3.1 Langevin equation as an SDE

In one dimension underdamped, the form of the Langevin equation for a random particle of mass m in a potential $V(x)$ is given by

$$m \frac{d^2x}{dt^2} = -\gamma \frac{dx}{dt} - V'(x(t)) + \sqrt{2\gamma k_B T} \zeta(t)$$

where

- γ is the friction coefficient
- $k_B T$ is the thermal energy
- $\zeta(t)$ is the white noise averaged to zero over time

The above equation can be simplified to observe the SDE structure

$$dx_t = v_t dt$$

$$mdv_t = \left(-\gamma v_t - V'(x(t)) \right) dt + \sqrt{2\gamma k_B T} dW_t$$

This is clearly an SDE form.

3.2 From Langevin SDE to Fokker-Planck equation

This is another central connection, for every Langevin SDE corresponds to a Fokker-Planck equation for a time changing probability density. For the underdamped Langevin SDE the associated Fokker-Planck equation is;

$$\frac{\partial P}{\partial t} = -v \frac{\partial P}{\partial x} + \frac{1}{m} \frac{\partial}{\partial v} [(\gamma v + V'(x(t)))P] + \frac{\gamma k_B T}{m^2} \frac{\partial^2 P}{\partial v^2}$$

The SDE describes individual paths, the Fokker-Planck describes the ensemble. [1]

3.3 How are SDEs, Langevin dynamics and generative modeling link?

Generative modeling is creating data from noise. SDEs, Langevin dynamics and generative modeling are linked through a single idea, that is, the usage of a stochastic differential equation to gradually transform an ordinary simple distribution into a complex data distribution. And then the goal is to learn the drift and score terms of that SDE from the data. This shows up as a diffusion model and score based generative models where the Langevin dynamics works as a sampling engine that runs the reverse time SDE. [4]

References

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- [4] Jasch Solh -Dickstein Yang Song. “SCORE-BASED GENERATIVE MODELING THROUGH STOCHASTIC DIFFERENTIAL EQUATIONS”. In: *arXiv:2011.13* (2021).